# Task 1 Neural Network

**Team ID:**

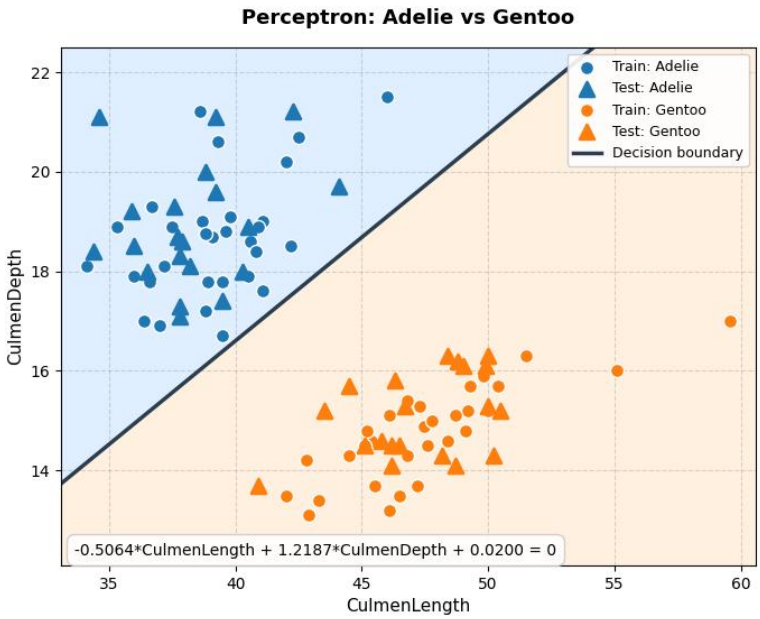
**TA:**

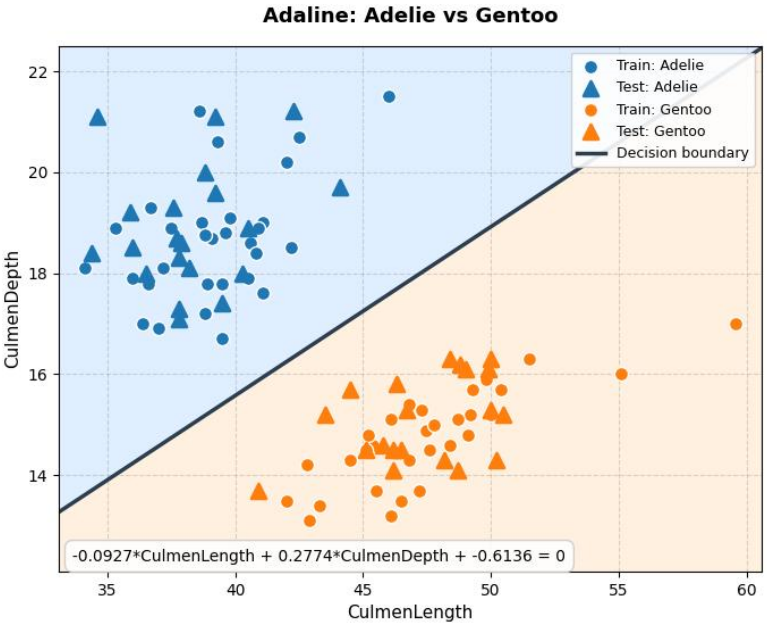
| Student Name | Student ID | Section number | Department |
|---|---|---|---|
| نورهان مدحت محمد عشري | 2023170682 | Section 6 | CS |
| يمني أيمن محمد بيومي | 2023170707 | Section 6 | CS |
| أمنية مصطفي جودة محمد | 2023170101 | Section 1 | CS |
| هنا مازن عبدالمولي محمد | 2023170690 | Section 6 | CS |
| يوسف محمد عاطف أحمد | 2023170739 | Section 4 | SC |

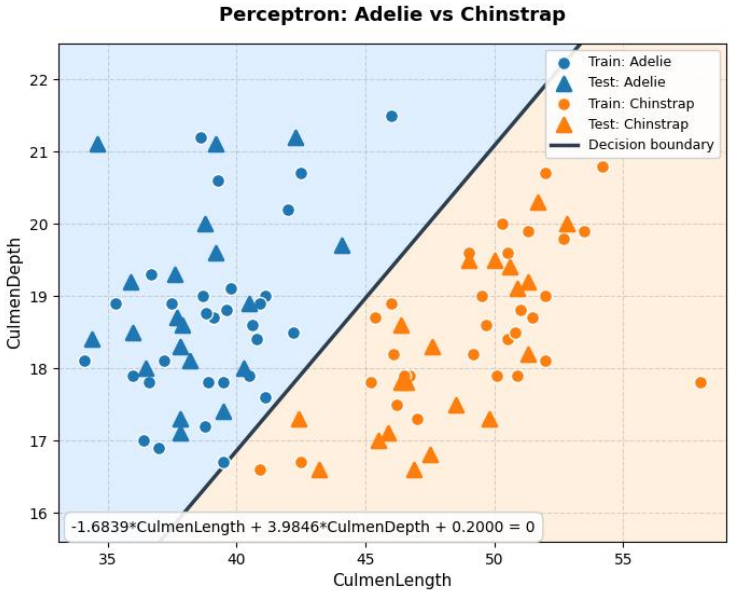
## Introduction

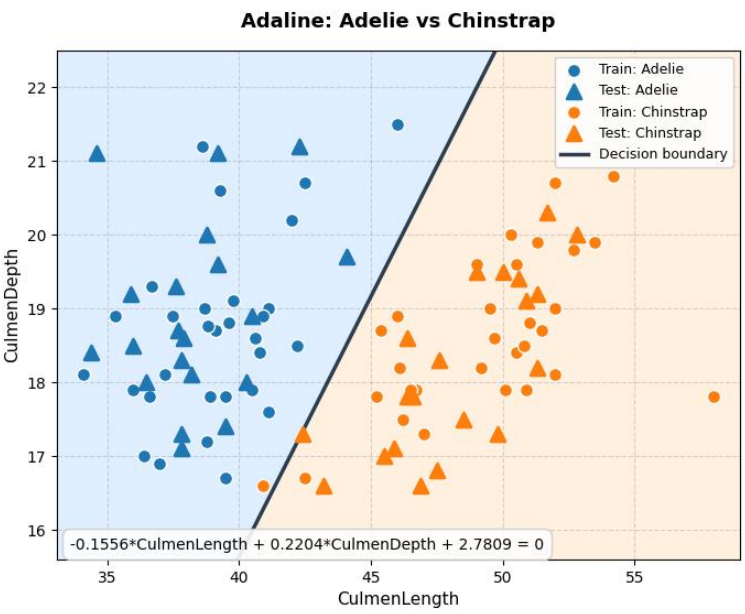
We are comparing Perceptron and Adaline using the penguins dataset. We picked 5 combinations of features to show what happens when data is easy (separable) and hard (overlapping)

## Good Separable Combinations



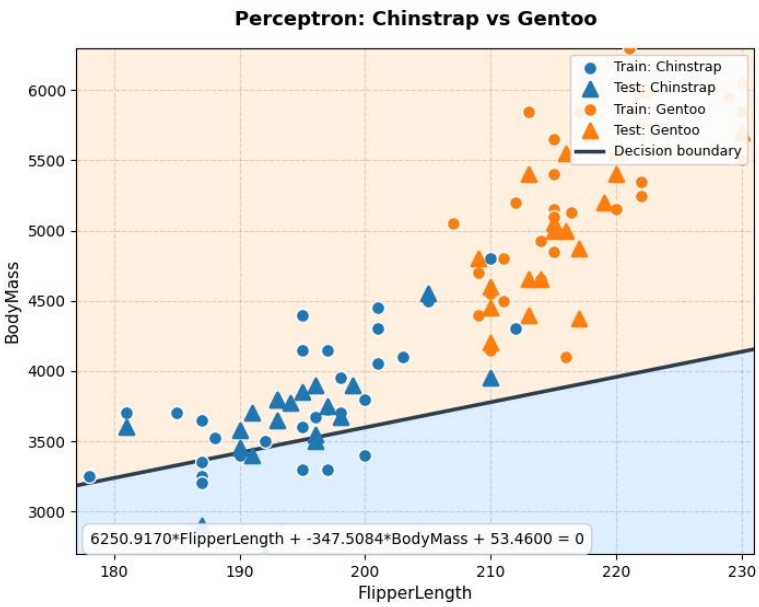



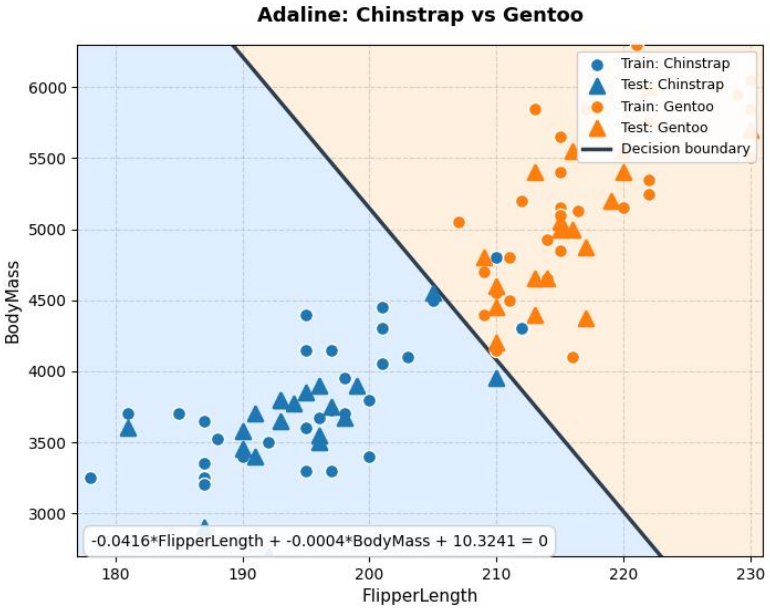




## Conclusion:

When the features give a clear gap between the penguins, both Perceptron and Adaline get 100% accuracy. They just draw a line in the empty space. This proves both algorithms work perfectly on linearly separable data

## Weird Case

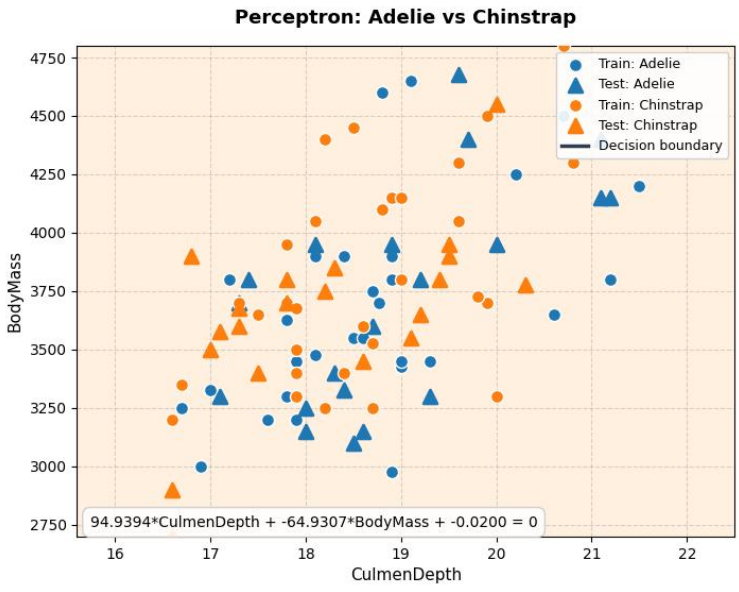


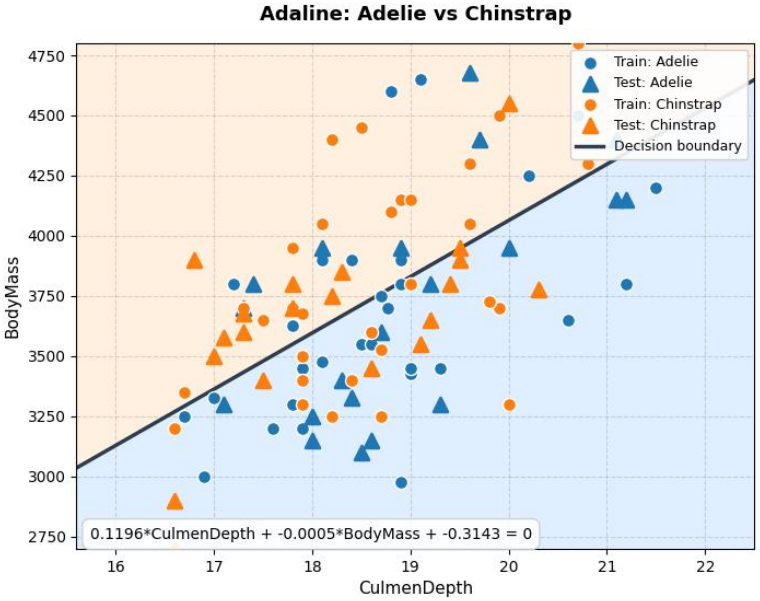



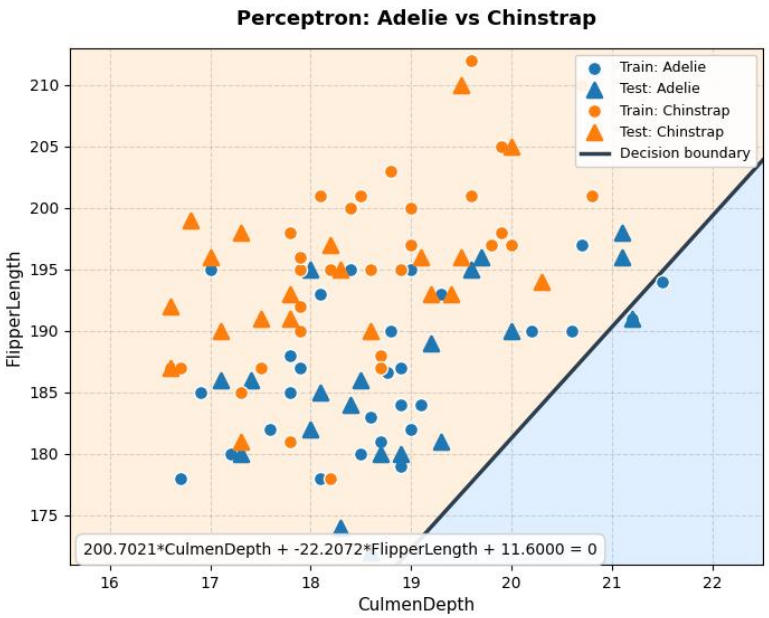

## Conclusion:

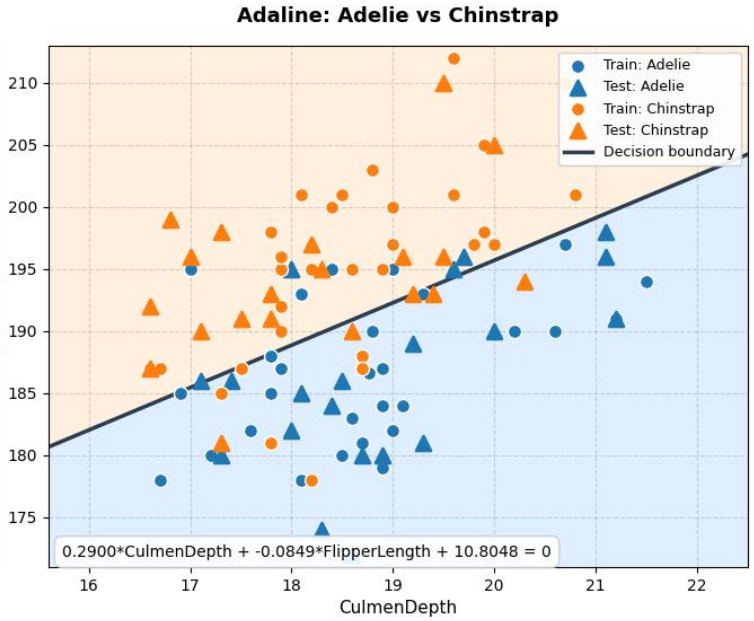
Even when data is separate, Perceptron sometimes gets stuck or draws a bad line if the learning rate or scales are off (it only got 60%). Adaline uses mean squared error to keep adjusting, so it found the perfect line (100%). This shows Adaline is more reliable

## Bad and Mediocre Separable Combinations






## Conclusion:

When the penguin features overlap, a straight line can't separate them. Perceptron totally freaks out because it never stops trying to find a perfect line that doesn't exist. Adaline is smarter because it just tries to find the "best possible" line with the lowest error, even if it's not perfect

## Table Results:

| Test Case | Classes | Features Used | Data Quality | Perceptron Accuracy | Adaline Accuracy |
|---|---|---|---|---|---|
| **Test 1** | Adelie vs Chinstrap | CulmenDepth & BodyMass | **Bad Overlap** | **50.0%** | **60.0%** |
| **Test 2** | Adelie vs Chinstrap | CulmenDepth & FlipperLength | **Mediocre Overlap** | **52.5%** | **80.0%** |
| **Test 3** | Adelie vs Gentoo | CulmenLength & CulmenDepth | **Good Separable** | **100.0%** | **100.0%** |
| **Test 4** | Adelie vs Chinstrap | CulmenLength & CulmenDepth | **Good Separable** | **100.0%** | **100.0%** |
| **Test 5** | Chinstrap vs Gentoo | FlipperLength & BodyMass | **Good Separable** | **60.0%** | **100.0%** |